ARTIFICIAL INTELLIGENCE IB ANNUAL CURRICULUM PROGRAM

Programme Name	LEVENT COLLEGE IB	Instructor Name	
Programme Duration (week)	40 weeks	Participant Name	
		Participant TR ID No	.

September 2024				
Week	**Subject**	**Objective**	**Practice / Assignment**	**Achievements**
1. Week	Course Introduction & What is AI?	History of AI, areas of use	AI example videos, ChatGPT experience	Recognizes AI concepts and explains different usage areas with examples.
2. Week	Python Installation & Environments	Python, VS Code, Jupyter Notebook setup	First "Hello World" code	Runs the first basic program using the Python environment.
3. Week	Python Basics 1	Variables, data types	Simple calculator program	Uses variables and data types for basic operations.
4. Week	Python Basics 2	Decision structures (if/else)	Average grade calculator code	Writes different codes using decision structures.

October 2024				
Week	**Subject**	**Objective**	**Practice / Assignment**	**Achievement**
5. Week	Loops	For and While loops	Code that finds a random number between 1–100	Automates repetitive operations using loops.
6. Week	Functions	Defining functions, parameters	Fahrenheit ↔ Celsius converter	Creates reusable modules by writing functions.

October 2024				
7. Week	Lists & Arrays	Create, add/remove list items	Classroom list application	Adds, deletes, and accesses items in lists.
8. Week	Dictionaries & Data Structures	Key-Value logic	Student information storage	Uses dictionary structures to store information.

November 2024				
Week	**Subject**	**Objective**	**Practice / Assignment**	**Achievement**
9. Week	File Reading/ Writing	TXT/CSV file operations	System that saves notes to a file	Opens, reads, writes, and saves files.
10. Week	Python Libraries	math, random, datetime	Number guessing game	Applies Python libraries (math, random, datetime).
11. Week	Introduction to NumPy	Array operations, mathematical calculations	Random data generation	Uses NumPy to process numerical data.
12. Week	Introduction to Pandas	Reading and filtering data	Student grades analysis with CSV	Processes and filters data with Pandas.

December 2024				
Week	**Subject**	**Objective**	**Practice / Assignment**	**Achievement**
13. Week	Introduction to Matplotlib	Plotting graphs	Classroom distribution chart	Visualizes data with Matplotlib.
14. Week	AI Logic	Basic machine learning concepts	Model, dataset concept	Explains the logic of AI and the model concept.
15. Week	Data Preprocessing	Handling missing data, normalization	Data cleaning with Pandas	Cleans datasets and applies normalization.
16. Week	Scikit-learn Introduction	Intro to simple ML algorithms	Installing scikit-learn	Runs simple ML models with scikit-learn.

January 2025				
Week	**Subject**	**Objective**	**Practice / Assignment**	**Achievement**
17. Week	Regression Analysis	Linear regression logic	House price prediction example	Explains and applies linear regression logic.
18. Week	Classification Algorithms 1	KNN algorithm	Handwriting recognition (MNIST subset)	Classifies data using the KNN algorithm.
19. Week	Classification Algorithms 2	Decision Tree, Random Forest	Fruit classification	Classifies with Decision Tree and Random Forest.
20. Week	Model Evaluation	Accuracy, precision, recall	Measuring model performance	Measures model accuracy and performance.

February 2025				
Week	**Subject**	**Objective**	**Practice / Assignment**	**Achievement**
21. Week	Image Processing + IoU	OpenCV basics	Image upload, grayscale, IoU calculation	Understands image processing and IoU/mAP concepts.
22. Week	Object Detection & Display	Pre-trained YOLO/SSD model	Dog-cat detection demo	Interprets outputs of pre-trained models like YOLO/SSD.
23. Week	Natural Language Processing (NLP) Intro	Text processing	Word counter app	Learns text processing and NLP basics.
24. Week	NLP & Simple Chatbot	if/else chatbot	Q&A chatbot system	Writes a basic chatbot.

March 2025				
Week	**Subject**	**Objective**	**Practice / Assignment**	**Achievement**
25. Week	AI Ethics & Safety	Data privacy, algorithm bias	Group discussion	Discusses AI ethics and safety issues.

March 2025				
26. Week	Project Preparation 1	Defining project scope	Selecting 3–5 projects, project plan	Defines project scope.
27. Week	Project Preparation 2	Data collection & labeling	LabelImg/Roboflow bounding box labeling	Collects data and performs labeling.
28. Week	Project Application 1	Transfer learning with YOLO/SSD	Training first model on Colab	Applies transfer learning with YOLO/SSD.

April 2025				
Week	**Subject**	**Objective**	**Practice / Assignment**	**Achievement**
29. Week	Project Application 2	Model evaluation	Calculating mAP50/mAP95	Evaluates models with mAP.
30. Week	Project Application 3	Improvement techniques	Data augmentation, parameter tuning	Improves model with data augmentation and tuning.
31. Week	Project Application 4	User interface development	Tkinter/Streamlit lite inference UI	Builds a basic user interface.
32. Week	Project Completion	Final tests and error fixing	Project file organization	Finalizes and tests project.

May 2025				
Week	**Subject**	**Objective**	**Practice / Assignment**	**Achievement**
33. Week	Project Presentation Preparation	Slide and report creation	PowerPoint/report writing	Prepares project report and slides.
34. Week	Project Presentations 1	Presentation skills	Group presentations	Presents project results.
35. Week	Project Presentations 2	Evaluating projects	Class feedback	Evaluates peers' projects.
36. Week	AI and the Future	AI's role in business world	Article analysis	Analyzes AI applications in real life.

June 2025				
Week	**Subject**	**Objective**	**Practice / Assignment**	**Achievement**
37. Week	Advanced Python Topics	Decorator, generator logic	Examples with small projects	Learns advanced topics in Python (decorator, generator).
38. Week	Deep Dive into AI Libraries	TensorFlow, PyTorch intro	Simple neural network	Runs a simple NN with TensorFlow/PyTorch.
39. Week	Year-End Review	Review of all topics	Short quiz and practice	Reviews yearly achievements.
40. Week	Year-End Evaluation	Portfolio submission	Project files	Submits project portfolio and receieves evaluation feedback.